

InuaBiz Architecture Specification

Cross-Platform Notification Engine & Multi-Role Alert Matrix

Comprehensive Blueprint for Real-Time In-App Bell Alerts, Push Notifications, WhatsApp/SMS Hooks & Transactional Emails

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Platform Stack:	React, Supabase PostgreSQL, Edge Functions	Document Version:	v1.0 — Technical System Architecture

1. Executive Summary & Strategy

A multi-tenant ecosystem like **InuaBiz** requires an instant, highly reliable notification dispatch system. Notifications bridge real-time user actions (e.g., customer purchases at a POS) with operational visibility (e.g., vendor low-stock warnings or super-admin subscription renewals).

Core Delivery Architecture

The notification engine utilizes **Supabase Realtime (WebSockets)** for zero-latency UI updates, combined with **Resend** for transactional emails and external SMS/WhatsApp API triggers for high-priority transaction confirmations.

2. Multi-Role Notification Matrix

Target Role	Event / Trigger	Channels Used	Operational Impact
Super Admin	New Vendor Registration	In-App Bell, Resend Email	Notifies admin of a new platform signup requiring verification or onboarding assistance.
Super Admin	SaaS Subscription Paid (KES 3,000)	In-App Bell, Resend Email	Confirms platform revenue generation and logs active subscription term.
Super Admin	Unclaimed Payment / Webhook Exception	In-App Bell, Urgency Banner	Alerts super-admin to manually map a payment whose <code>api_ref</code> failed to match a tenant.
Vendor (Store Owner)	New Sale / Order Completed	In-App Audio Ping + Bell, Push Notification	Provides instant POS payment confirmation during live customer transactions.
Vendor (Store Owner)	Low Stock Threshold Reached	In-App Bell, Daily Email Summary	Triggers inventory restock alerts when product quantities fall below minimum safety levels.
Vendor (Store Owner)	Subscription Due / Expiry Alert	In-App Modal, WhatsApp / SMS, Email	Warns vendor 3 days prior to trial/subscription expiration to ensure uninterrupted access.
End Customer			

Target Role	Event / Trigger	Channels Used	Operational Impact
	Digital Receipt / STK Push Prompt	M-Pesa Screen Prompt, SMS / Email Receipt	Delivers instant payment validation and digital proof of purchase following a store sale.

3. Multi-Channel Tech Stack Specification

- **In-App Bell & Badge (Supabase Realtime):** Subscribes directly to PostgreSQL Row-Level Changes on the notifications table. Updates count badges instantly without page refresh.
- **Transactional Email (Resend API):** Handles HTML-styled branded emails for receipts, subscription invoices, password resets, and admin summaries.
- **SMS & WhatsApp Webhooks (Infobip / Africa's Talking / Twitch):** Dispatches critical time-sensitive alerts (e.g., M-Pesa failure alerts or subscription warnings) directly to vendor phones.
- **Sound Audio Cues (Web Audio API):** Triggers an audible chime on the vendor POS page whenever a payment completes successfully.

4. System Architecture & Trigger Pipeline

```
[ Action Event ] (e.g. Sale / Payment / Low Stock) | ▼ [ Supabase PostgreSQL DB Trigger / Edge Function ] | |→ (1. INSERT into `notifications` table) → [ Supabase Realtime WebSocket ] → [ Frontend In-App Bell ] | |→ (2. High Priority Email?) → [ Resend API ] → [ User Inbox ] | |→ (3. High Urgency SMS/WhatsApp?) → [ External SMS Gateway ] → [ User Mobile Phone ]
```

5. Database Schema: Unified Notifications Table

All notifications across the platform are stored in a centralized, highly indexable notifications table using Row Level Security (RLS) to enforce data privacy.

Column Name	Data Type	Constraints	Description
id	UUID	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique notification ID.
tenant_id	UUID	NULLABLE (NULL for Super Admin)	Isolated to specific vendor account.
recipient_role	VARCHAR(30)	NOT NULL	'SUPER_ADMIN', 'VENDOR_ADMIN', or 'CUSTOMER'.
recipient_id	UUID	NOT NULL, FOREIGN KEY -> profiles(id)	Target user account ID.
title	VARCHAR(150)	NOT NULL	Short header (e.g., "New Sale Completed").
message	TEXT	NOT NULL	Detailed alert content.
type	VARCHAR(50)	NOT NULL	'SALE', 'STOCK_LOW', 'SUBSCRIPTION', 'SYSTEM'.

Column Name	Data Type	Constraints	Description
priority	VARCHAR(20)	DEFAULT 'NORMAL'	'LOW', 'NORMAL', 'HIGH', 'CRITICAL'.
is_read	BOOLEAN	DEFAULT FALSE	Tracks unread state for badge counter.
metadata	JSONB	NULLABLE	Stores contextual payload (e.g., { "amount": 1500, "sale_id": "..."}).
created_at	TIMESTAMPTZ	DEFAULT NOW()	Event timestamp.

6. Delivery Fallback & Notification Preferences

- **Role-Based Row Level Security (RLS):** Super admins can view platform-wide system alerts, while vendors can only query notifications where `tenant_id = auth.jwt() -> tenant_id`.
- **Batch Notification Triggers:** Non-critical alerts (like daily stock digest) are batched via scheduled background jobs to prevent spamming users.
- **User Preference Center:** Vendors can customize their channel preferences (e.g., enable/disable email receipts while keeping sound alerts enabled).